

American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

E-Farm Management System

A Software Engineering Project Submitted
By

Semester: Spring_24_25		**Section: U**	**Group Number: 05**	
SN	Student Name	Student ID	Contribution (CO3+CO4)	Individual Marks
1.	Shahadat Hossain	23-51341-1	35%	
2.	Hamidul Islam	23-50447-1	35%	
3.	Mostafizur Rahman	23-50829-1	30%	

The project will be evaluated for the following Course Outcomes

CO3: *Select* appropriate software engineering models, project management roles, and their associated skills for the complex software engineering project and evaluate the sustainability of developed software, taking into consideration the societal and environmental aspects	Total Marks	
Appropriate Process Model Selection and Argumentation with Evidence	[5 Marks]	
Evidence of Argumentation Regarding Process Model Selection	[5Marks]	
Analysis of the impact of societal, health, safety, legal, and cultural issues	[5Marks]	
Submission, Defense, Completeness, Spelling, grammar, and Organization of the Project report	[5Marks]	
CO4: *Develop* a project management plan to manage software engineering projects following the principles of engineering management and economic decision process	Total Marks	
Develop the project plan, its components of the proposed software products	[5Marks]	
Identify all the activities/tasks related to project management and categorize them within the WBS structure. Perform detailed effort estimation correspond with the WBS and schedule the activities with resources	[5Marks]	
Identify all the potential risks in your project and prioritize them to overcome these risk factors.	[5Marks]	

CO5: Perform as an effective team member or leader in diverse team settings and solve multi-disciplinary problems in the computer science and engineering domain	Total Marks	
Taking project responsibility: perform assigned tasks on time independently	[5 Marks]	
Contribution to project group meetings, sharing fruitful ideas	[5Marks]	
Positive attitude towards group work, collaboration, compromise, helping others to understand their project work responsibility	[5Marks]	
Showing respect and value towards other team member's opinion	[5Marks]	

Description of Student's Contribution in the Project work

Student Name: Shahadat Hossain
Student ID: 23-51341-1
Contribution in Percentage (%): 35%
Contribution in the Project:

- Contribution Description 1:
    1. Project Proposal:
        - Background to the Problem
        - Solution to the Problem
    2. Literature Review
    3. Process Model
    4. Project Roles and Responsibilities
    5. Requirements analysis:
        - Functional Requirements
        - Non-functional Requirements

- Contribution Description 2:
    1. Effort Estimation:
        - COCOMO Calculation
        - EVA Calculation
    2. Risk Management



Signature of the Student

Student Name: Hamidul Islam
Student ID: 23-50447-1
Contribution in Percentage (%): 35%
Contribution in the Project:

- Contribution Description 1:
    1. Project Proposal
    2. UI/UX Design
    3. Class Diagram

4. Use Case Diagram

- Contribution Description 2:
    1. Test Case
    2. WBS
    3. Gantt Chart

Hasib

_____
Signature of the Student

Student Name: Mostafizur Rahman
Student ID: 23-50829-1
Contribution in Percentage (%): 30%
Contribution in the Project:

- Contribution Description 1:
    1. Use Case Diagram
    2. Sequence Diagram
- Contribution Description 2:
    1. Test Case
    2. Gantt Chart

Mostafizur

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Signature of the Student

E-Farm Management System

1. PROJECT PROPOSAL

1.1 Background to the Problem

Agriculture forms the backbone of Bangladesh's economy and plays a pivotal role in the livelihoods of millions of people across the country. A large portion of the rural population depends directly or indirectly on farming to sustain their families. From rice paddies in the northern plains to vegetable farms in the central and southern regions, Bangladesh's food supply heavily relies on small-scale farmers. Despite this massive agricultural activity, the sector is plagued by a few inefficiencies and systemic problems. Farmers often struggle to receive fair compensation for their produce due to the influence of middlemen, lack of access to proper markets, and limited government support mechanisms. Additionally, there is no unified platform or infrastructure that allows the government to track crop production, distribute resources, or support farmers in a timely and structured manner.

In many cases, even after a successful harvest, farmers are left with unsold stocks or are forced to sell their goods at a much lower price than the market rate because they lack the bargaining power or access to a wider market. The absence of a transparent and farmer-friendly supply chain results in price volatility, artificial scarcity in urban areas, and substantial post-harvest losses. Simultaneously, farmers who require loans, subsidies, seeds, or farming equipment often face bureaucratic delays or don't even apply due to a lack of awareness or access to the proper channels. Moreover, government agencies do not have real-time or accurate data on what types of crops are being grown, where surpluses or shortages exist, and how to balance demand across different regions.

The **E-Farm Management System** has been envisioned as a digital platform developed under a government-led initiative to address these widespread issues in a holistic and sustainable way. This software, designed with object-oriented programming principles, aims to provide a seamless interface between the government and the country's farming population. It allows government officials (admins) to create, manage, and monitor farmer profiles, collect and track data on crop production, handle distribution logistics, and ensure equitable access to resources such as loans, seeds, and farming equipment. At the same time, farmers can log into applications either through a smartphone or with help from government employees—to request to sell their stock, apply for subsidies, rent machinery, or update their production records.

By bringing all these services under one unified platform, **the E-Farm Management System** bridges the gap between policy and practice. It not only empowers farmers with better access to government services but also enhances transparency, reduces dependency on middlemen, and improves the government's ability to formulate evidence-based policies using real-time data. This system ultimately helps increase agricultural productivity, stabilize prices, and ensure food security across all regions of Bangladesh.

Root Cause of the Problem

The challenges faced by farmers in Bangladesh stem from several deep-rooted structural, economic, and logistical issues. These challenges, if left unaddressed, continue to perpetuate poverty, food insecurity, and agricultural inefficiency. The core problems include:

- **Absence of Centralized Agricultural Data**: Currently, there is no unified government platform that records the monthly production, type, and quantity of crops being cultivated in different regions. This lack of data severely limits the government's ability to plan stock redistribution, regulate prices, or provide targeted support.
- **Over-Reliance on Intermediaries and Middlemen**: Most farmers depend on third-party agents or middlemen to sell their produce. These intermediaries often exploit the lack of transparency by purchasing goods at artificially low prices and reselling them at high margins in urban markets. As a result, farmers rarely receive a fair share of the market value.

- **Inefficient Resource Allocation and Support System**: Farmers often do not receive seeds, fertilizers, or government-subsidized materials on time, and there is no formal process for requesting or tracking these needs digitally. This results in low yield and hampers production potential.
- **Complexity in Accessing Financial Aid and Subsidies**: The process for applying for farming loans or subsidies is outdated and full of red tape. Most farmers are unaware of the available schemes or are unable to access them due to lack of documentation, digital literacy, or assistance.
- **Lack of Real-Time Monitoring and Delivery Management**: When farmers attempt to sell their harvests, there is no real-time system to track demand in different regions or arrange for transport efficiently. This causes produce to rot in storage or leads to unbalanced food availability across the country.
- **Technology Access and Digital Literacy Gap**: Many farmers still rely on feature phones and may not be comfortable using mobile applications. Without support from local government agencies, a purely digital solution would not be accessible to all. Hence, the system must be admin-assisted.

Importance of Solving this Problem

Solving the problems faced by farmers is critical not only for individual welfare but for the broader economic and social development of the country. The **E-Farm Management System** offers a range of solutions that directly contribute to this goal:

1. **Empowering Farmers Economically**: By enabling farmers to submit sale requests and negotiate directly through the platform, the software removes middlemen and ensures fairer prices, improving the economic condition of farming communities.
2. **Efficient Government Planning**: The app collects valuable data such as crop type, expected harvest dates, stock availability, and sale records. This data helps policymakers analyze trends, predict shortages, manage regional distribution, and set appropriate price floors or ceilings.
3. **Faster and Fair Access to Subsidies and Loans**: A transparent, trackable digital process ensures that financial assistance reaches deserving farmers without corruption or delay. Farmers can easily apply and reapply with corrected documentation if needed.
4. **Resource Optimization**: By centralizing requests for seeds, fertilizers, and farming machinery, the government can optimize its stock management and ensure that resources reach those who need them the most, when they need them.

5. **Data-Driven Policy Making**: With updated regional production and market data available to government agencies, it becomes easier to develop fair pricing policies, design support schemes, and forecast food security challenges.
6. **Boosting National Food Security**: Through strategic redistribution of farming products based on demand in districts like Dhaka, Chattogram, or Sylhet, the system minimizes wastage and ensures continuous food supply in all areas.
7. **Creating an Incentive Structure**: The platform introduces a "priority level" system (Bronze to Platinum) where farmers gain rewards and faster service access by actively updating their production records and participating in the system. This motivates users to stay engaged.
8. **Inclusive Ecosystem for All Stakeholders**: The system also integrates transport service providers by allowing verified drivers to sign up and get assigned deliveries. This not only supports timely logistics but creates employment opportunities across sectors.

By providing a streamlined, end-to-end solution, the **E-Farm Management System** has the potential to revolutionize agricultural management in Bangladesh. It brings transparency, accountability, and accessibility to one of the most critical sectors of the country's economy. Most importantly, it ensures that the people who work hardest to feed the nation are finally empowered with the tools they need to thrive.

1.2 Solution to the Problem

Project Objectives

The E-Farm Management System is developed with the following key objectives in mind:

- **Centralization of Agricultural Data:** To create a unified, government-managed platform that records crop production, regional output, and farmer activity across the country, enabling better planning and monitoring.
- **Reduction of Middleman Exploitation:** To empower farmers by allowing them to directly submit stock for sale to the government, thereby removing third-party intermediaries and ensuring fair pricing.
- **Streamlining Resource Distribution:** To digitize and simplify the process of requesting and delivering seeds, fertilizers, machinery, and other farming necessities on time, thus boosting productivity.

- **Facilitation of Financial Aid:** To provide an easy-to-use digital process for farmers to apply for loans and subsidies, reducing bureaucratic delay and increasing awareness of government programs.
- **Improvement of Transparency and Logistics:** To enable real-time monitoring of crop supply, regional demand, and transportation logistics, ensuring smoother delivery and reduced waste.
- **Encouragement of Farmer Participation:** To introduce an incentive-based ranking system that motivates farmers to remain engaged and regularly update their profiles and production records.

Proposed Solution

The proposed solution to Bangladesh's systemic agricultural challenges is a digital government-backed platform named **E-Farm Management System**, designed using object-oriented programming techniques. This software will serve as a bridge between farmers and various governmental services to address inefficiencies and promote transparency across the agricultural sector. Farmers can access the platform via smartphones or with assistance from local government staff, allowing them to:

- Register or update their agricultural profiles
- Submit requests to sell their crops directly
- Apply for subsidies, loans, or farming equipment
- Update seasonal production records

Meanwhile, government administrators can:

- View, manage, and verify farmer data
- Track regional production trends
- Allocate resources more effectively
- Organize transport and logistics using registered delivery drivers

A reward-based **"Farmer Priority Level System"** (ranging from Bronze to Platinum) will encourage farmers to remain active and accurate in their reporting. The system integrates smart matching of surplus to demand zones (e.g., Dhaka, Sylhet, Chattogram) and assigns verified delivery drivers for efficient redistribution.

This solution is particularly appropriate as it:

- Leverages existing digital infrastructure and government networks
- Reduces the digital literacy barrier via admin-assisted interfaces
- Aligns directly with national goals of food security and economic empowerment
- Is scalable and adaptable to various regions and farming scales

The feasibility of this solution has been validated through prototype testing and aligns with business goals of sustainable agriculture, reduced waste, and economic upliftment.

Basic Functionalities

The E-Farm Management System consists of several core features that utilize modern technology and deliver real-world societal impact:

- **Farmer/Admin Profiles:** Role-specific dashboards allow farmers to submit requests and receive support, while admins manage approvals and monitor data trends.
- **Digital Resource Request & Allocation:** Seamlessly process requests for seeds, loans, and farming equipment, reducing bureaucracy.
- **Smart Stock Submission:** Farmers can offer stock for sale, view government purchase offers, and track sales history.
- **Real-Time Monitoring & Demand Matching:** The system tracks where food is needed, preventing stock wastage and managing logistics.
- **Transport Coordination:** Verified drivers can register and be auto-assigned for deliveries, reducing delays in redistribution.
- **Priority Incentive System:** Farmers who maintain accurate, timely data entries are rewarded with faster service and recognition.
- **Security and Transparency:** Government-level data encryption and audit logs ensure all user activity is traceable and protected.

Target Users and Benefits

Primary Users:

- **Small and Medium Farmers:**
    - Fair crop prices through direct government sales
    - Easier access to subsidies and equipment

    - Better market awareness and reduced waste
- **Government Administrators:**
    - Real-time data for policy planning
    - Efficient distribution of agricultural resources
    - Transparency in fund and supply management
- **Transport Service Providers (Drivers):**
    - New job opportunities via verified logistics coordination
    - Assigned tasks based on location, availability, and trust score

Indirect Beneficiaries:

- **Consumers and Urban Populations:**
    - Stable food prices and availability
    - Reduced artificial scarcity
- **Policy Makers and Economists:**
    - Access to analytical data for food security decisions

Contribution of the Project

This project contributes significantly to technological and social advancement by integrating:

- **Data Science and Predictive Analysis:** Supports forecasting food demand and crop yield trends
- **Object-Oriented Software Design:** Offers scalable, modular, and maintainable architecture
- **IoT Integration Potential:** Can later integrate sensors or devices for crop health monitoring
- **Public Sector Efficiency:** Digitizes manual processes, saving time and resources
- **Behavioral Insights:** Uses gamification (priority levels) to encourage proactive user behavior

It lays a foundation for broader scientific studies in food logistics, e-governance, and socio-economic upliftment, while offering a replicable model for other developing nations.

Literature Review

Recent research in agricultural software solutions has focused on digitizing rural farming processes. For instance, studies by Islam et al. (2019) and Rahman & Khatun (2021) explored the need for centralized farming databases in Bangladesh but lacked implementation strategies. Platforms like **Krishi Call Centre** and **Digital Krishok** offer farmer support but are limited to advisory roles rather than logistics and sales management.

Comparatively, global solutions like **AgriDigital** (Australia) and **e-Soko** (Ghana) provide market data and trading platforms, but their regional applicability to Bangladesh remains limited due to infrastructure and policy differences.

The E-Farm Management System extends these efforts by combining government integration, real-time logistics, smart supply matching, and inclusive access—all tailored to the socio-economic landscape of Bangladesh.

Existing Software Solutions

Several platforms currently attempt to tackle agricultural inefficiencies:

- **Krishi Call Centre (Bangladesh):** Offers telephonic advice to farmers but lacks logistics or subsidy management.
- **e-Krishi:** Facilitates market price info but doesn't enable stock submission or delivery coordination.
- **Digital Krishok:** Focuses on knowledge sharing, not financial aid or resource allocation.

How E-Farm Management System Extends These:

- **End-to-End Functionality:** From farmer registration to delivery logistics
- **Priority System and Incentivization:** Encourages sustained user engagement
- **Integrated Logistics and Verified Drivers:** Unique to this system
- **Governmental Role in Real-Time Operations:** Enables immediate data use for policymaking

With these improvements, the E-Farm Management System offers a powerful, scalable solution tailored to Bangladesh's agricultural and digital evolution.

References:

1. **Islam et al. (2019) – Centralized Farming Databases in Bangladesh**
   Study on climate adaptation and the importance of data systems for farmers in Bangladesh.
   Link: https://link.springer.com/article/10.1007/s10113-022-01962-8

2. **Krishi Call Centre – Advisory Services for Farmers**
   Bangladesh government initiative providing agricultural advice via call center.
   Link:
   https://www.dhakatribune.com/bangladesh/agriculture/345509/farmers%E2%80%99-friend-krishi-call-centre-struggling-to

3. **Digital Krishok (e-Krishok) – ICT Agricultural Platform**
   An ICT-based solution offering market linkage and farming information to Bangladeshi farmers.
   Link: https://comdevasia.org/rural-communication/e-krishok-a-360-degree-ict-enabled-solution-to-empower-farmers-in-bangladesh

4. **AgriDigital (Australia) – Digital Grain Supply Chain**
   Global example of a digital platform for grain management and supply chain digitization.
   Link: https://www.agridigital.io

5. **e-Soko (Ghana) – Market Information and Trading Platform**
   Agricultural data and trading service for farmers in Ghana.
   Link: https://www.esoko.com

2.1 Process Model:

The proposed software project, **E-Farm Management System**, aims to address widespread inefficiencies in Bangladesh's agricultural sector through a centralized, government-led digital platform. The system enables farmers to request loans, register crops, apply for subsidies, and manage logistics—while government officials handle real-time data, distribution, and support mechanisms. After careful analysis of the project's scope, complexity, user diversity, and evolving requirements, a **hybrid approach combining the Incremental Development Model and Scrum (Agile)** is selected as the most suitable process model.

Incremental Development + Scrum Overview:

- The **Incremental Model** breaks down the system into manageable, independent modules that can be developed, tested, and delivered step by step.
- **Scrum**, an Agile framework, is integrated during the implementation phase to manage development through time-boxed sprints, with continuous feedback and iterative improvements.

This hybrid model offers both **structured planning and flexible delivery**, making it ideal for academic projects with both technical and reporting milestones.

Applicability to E-Farm Management System:

- **Modular and Complex Requirements:** The system includes various modules such as farmer registration, stock sale, financial aid application, production tracking, and driver logistics. Each of these modules can be treated as an increment, developed in separate sprints under Scrum, ensuring timely progress and parallel development.
- **Dynamic Stakeholder Environment:** The target users like farmers, government admins, and transport drivers have different expectations and technology access levels. Scrum enables iterative development and user feedback after each sprint to accommodate changing user needs and system improvements.

Comparison with Other Models:

1. **Waterfall Model**
   Not Used Because: The Waterfall model is too rigid for a system that demands ongoing feedback and modular releases. Once requirements are set in Waterfall, changes are difficult, which conflicts with the real-world variability of agricultural policies and user behaviors.
2. **Prototyping Model**
   N*ot Used Because*: While prototyping is useful for UI validation, it doesn't provide the structured delivery and scalability required for a system with many interconnected components.
3. **V-Model**
   Not Used Because: This model adds intensive testing at each step, which can be overkill for a small team working on a limited timeline. It also lacks the flexibility to respond to new ideas or user input.

4. **Component-Based Development**
   Not Used Because: This model is ideal for reusing existing components, but the E-Farm System demands highly customized functionality tailored to local government workflows and regional farming needs, which are unlikely to be available as pre-built components.

5. **Rapid Application Development (RAD)**
   Not Used Because: While RAD emphasizes speed, it downplays long-term structure. Given the diverse and critical user base (government, rural farmers), more planning and structured increments are needed than RAD alone offers.

6. **Extreme Programming (XP), DSDM, FDD**
   Not Used Because: These models require high developer discipline, frequent releases, and deep stakeholder involvement all of which may not be feasible for a public-sector software project.

Advantages of Using Incremental and Scrum:

- **Parallel Development**: Teams can work on different modules independently, Examples: login system, loan requests, delivery assignments.
- **Early Delivery of Core Features**: Registration, stock sale requests, and basic admin dashboards can be deployed early and expanded later.
- **Flexible for Feedback**: Scrum enables mid-development input from users or faculty, which is crucial for improving usability and accuracy.
- **Time-Boxed Sprints**: Allows well-organized planning and time management for each semester milestone.
- **Risk Reduction**: Smaller increments reduce the chance of full project failure. Bugs or feedback can be contained and addressed sprint by sprint.
- **Continuous Feedback Loop**: Regular sprint reviews encourage feedback from instructors and potential stakeholders (e.g., simulated farmers/admins), helping the team refine features like loan approval, priority ranking, and subsidy tracking in real time.
- **Supports Diverse User Needs**: Farmers, government admins, and drivers have different digital skills and expectations. Incremental development allows tailoring of interfaces and workflows specific to each group, promoting better adoption.
- **Improved User-Centered Design**: Frequent feedback cycles in Scrum allow the UX/UI team to iterate rapidly and enhance usability for semi-literate users by incorporating local language support, icons, and offline capabilities.

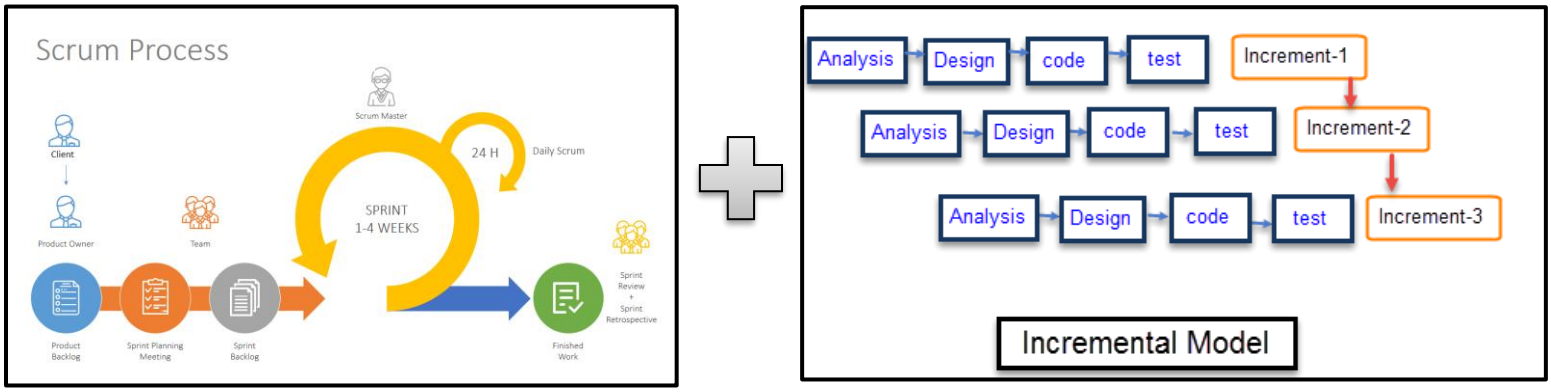
- **Low Risk of Misalignment**: Since each sprint includes a review and demonstration, there's a minimal chance of building a system that deviates from the actual needs. This continuous alignment is something traditional models like Waterfall often lack.
- **Adaptability to Policy Changes**: Government processes or agricultural policies may change during the semester. Scrum allows quick re-prioritization of the product backlog, enabling the development team to shift focus without derailing the whole project.
- **Faster Issue Detection and Resolution**: Because testing is embedded in each increment, bugs or performance issues are identified and resolved sooner, improving the reliability and security of modules like OTP login or subsidy validation.
- **Scalability for Future Enhancements**: Features like weather-based alerts, IoT device integration, or AI-driven trend analysis can be developed in future increments without disrupting the existing system, a clear advantage over rigid models.
- **Better Collaboration and Ownership**: Scrum's daily stand-ups, sprint planning, and retrospective meetings promote strong communication and shared accountability within the team, boosting productivity and problem-solving.

Implementation Plan Using the Incremental + Scrum Model:

1. **Requirement Analysis & Planning (Week 1–2)**
    - **Define System Structure:** Define functional requirements, use cases, and WBS
    - **Module Prioritization:** Prioritize modules: Start with core features like registration and dashboard.
2. **Sprint 1 (Week 3–4):**
    - **User Access Implementation:** Farmer registration, login system, and admin approval interface.
    - **Sprint Evaluation:** Sprint review and feedback.
3. **Sprint 2 (Week 5–6):**
    - **Core Functional Modules:** Stock sale requests, production updates, and real-time dashboard.
    - **Quality Assurance:** Testing and improvement based on demo.
4. **Sprint 3 (Week 7–8):**
    - **Application Features:** Loan/subsidy application, raw material request module.
    - **Workflow Enhancements:** More complex admin workflows and integration logic.

5. **Sprint 4 (Week 9–10):**
    - **Extended User Roles & Services:** Delivery driver registration, government lease application, reporting system.
    - **System Validation:** Final system testing and bug fixes.
6. **Sprint 5 (Week 11–12):**
    - **System Finalization:** UI polishing, security audit, offline mode implementation.
    - **Project Closure:** Prepare for final defense and submission.

Therefore, by adopting a hybrid **Incremental + Scrum** model, the E-Farm Management System benefits from a structured roadmap and adaptive development. It supports modular deployment, incorporates feedback loops, and ensures the system is reliable, user-friendly, and aligned with real-world agricultural needs. This model is superior to traditional and rigid methodologies in the context of an academic, multi-module software project with dynamic user needs.



2.2 Project Roles and Responsibilities:

1. **Product Owner (Government Project Coordinator / Instructor Proxy)**
    - Defines the project goals and ensures that the system aligns with national agricultural priorities and stakeholders' needs.
    - Manages the feature backlog including modules like farmer registration, stock submission, loan applications, and driver assignments.
    - Works closely with the development team to prioritize tasks based on impact and urgency and ensures that sprint deliverables support the overall vision of the E-Farm Management System.

2. **Scrum Master**
    - Facilitates daily stand-ups, sprint planning, and retrospectives to keep the team aligned and focused.
    - Removes blockers such as delays in receiving requirements, UI feedback, or clarification on backend logic.
    - Ensures that Agile principles are followed during sprint execution, while integrating elements of structured planning from the Incremental model.
3. **Development Team**
    - Responsible for designing, coding, testing, and integrating all core functionalities, such as the farmer dashboard, loan request workflows, admin verification systems, and driver coordination tools.
    - Develops user interfaces optimized for both web and mobile platforms, ensuring support for multilingual and offline access features.
    - Implements secure backend processes including NID verification, OTP validation, and encrypted data management.
    - Conducts continuous integration and unit/system testing after each sprint to ensure reliability and performance.
4. **Stakeholders (Farmers, Government Admins, Delivery Drivers)**
    - Provide critical user feedback during usability reviews, particularly on areas like the loan application process, priority ranking system, and resource request functionality.
    - Collaborate with the product owner to validate whether the system meets real-world expectations, especially in regions with low digital literacy or network infrastructure.
    - Highlight feature gaps, process bottlenecks, or interface challenges for iterative improvements.
5. **Project Manager**
    - Oversee the project timeline and resource allocation, ensuring that milestones are met within academic deadlines.
    - Coordinates with all roles to ensure balanced workloads and timely task completion across sprints and development phases.

- Addresses risks related to technology limitations, design delays, or integration conflicts, and ensures documentation aligns with academic requirements.

6. **Quality Assurance (QA) Team**
    - Validates core features such as the password recovery mechanism, subsidy request verification, and admin-farmer communication channels for correctness and reliability.
    - Executes structured testing: unit testing, integration testing, and user acceptance testing before each sprint review or academic submission.
    - Ensures compliance with security protocols and data handling standards mandated by local regulations.
7. **UI/UX Designer**
    - Designs user-friendly and accessible interfaces, using visual cues and Bengali localization to support semi-literate users in rural areas.
    - Creates responsive and intuitive dashboards for both farmers and admins, ensuring clarity in all user interactions.
    - Conducts user experience testing sessions to improve navigation, minimize cognitive load, and maximize engagement, particularly for features like the reward tier system and resource request forms.
8. **Support and Maintenance Team**
    - Manages post-sprint feedback, bug reports, and system logs for issues such as request failures or driver assignment errors.
    - Provides basic system help documentation, guides, and in-app assistance features (e.g., tooltips, status trackers).
    - Prepares for potential enhancements such as IoT integration or weather forecasting modules for future scalability and maintenance planning.

3. REQUIREMENTS ANALYSIS

3.1. Functional Requirements

Farmer Registration and Account Management

1. Admins can register new farmers by entering required details: Full name, NID, phone number, address, and farming type.
2. The system verifies the NID and phone number to ensure authenticity.
3. Farmers receive credentials (username and password) to access the system.
4. Admins can update or delete farmer accounts when necessary.

Priority: High

Farmer Login

1. Farmers can log in with registered phone number and password.
2. System verifies credentials with database.
3. On successful login, the farmer dashboard will be displayed.
4. If login fails 5 times consecutively, the account will be temporarily locked for 1 hour.
5. Farmers can recover passwords using a registered phone number or NID.

Priority: High

Recover Password

1. Farmers provide their NID or registered phone number to reset password.
2. The system checks if the provided data exists in the database.
3. A 6-digit OTP will be sent to the verified contact.
4. Farmers must verify OTP before setting a new password.
5. Passwords must follow a strong password policy (uppercase, lowercase, number, special character).
6. A confirmation SMS will be sent upon successful reset.

Priority: Medium

Stock Sale Request Management

1. Farmers can request to sell their stock by entering product type, quantity, price, and harvest date.
2. Admins validate and approve/reject requests based on market demand and logistics.
3. If approved, the system schedules pickup and tracks product delivery.
4. Farmers can cancel or edit sale requests before admin approval.

Priority: High

Loan and Subsidy Application

1. Farmers can apply for financial assistance by providing bank details, land info, crop type, and desired amount.
2. Admins validate and forward application for on-site inspection.
3. Upon approval, the government processes the fund transfer to the farmer's bank account.
4. Farmers can update and resubmit denied applications.

Priority: High

Raw Material and Machinery Request

1. Farmers can request raw materials (fertilizers, seeds) or rent machinery through the system.
2. Admins review the request and approve or deny based on availability.
3. Subsidized rates apply for government-approved users.
4. Machine rentals are processed monthly and logged in farmer history.

Priority: High

Production and Sale Update

1. Farmers can record harvested product details: type, date, quantity, price, and market.
2. Admins verify entries to maintain record accuracy.
3. The system logs valid contributions and calculates farmer activity score.

Priority: Medium

Farmer Priority System

1. Farmers are ranked as Bronze, Silver, Gold, or Platinum based on transparency, activity, and valid data submission.
2. Higher-tier farmers receive prioritized services like faster loan approvals or product approvals.
3. Tier score is visible on the farmer dashboard.

Priority: Medium

Government Land or Pond Lease Application

1. Farmers can apply to lease government-owned properties like ponds or land.
2. Admins review and assign an investigator for physical verification.
3. Lease approvals are recorded, and the farmer gets an official lease certificate.

Priority: Medium

Driver Registration and Delivery Assignment

1. Drivers can register using NID, license, phone number, and vehicle details.
2. Admins validate and approve registered drivers.
3. Admins assign delivery tasks based on availability and destination.
4. Drivers are paid wages with government incentives based on distance and load.

Priority: High

3.2. Non-functional Requirements

Usability

1. The system should have a simple UI supporting local language (e.g., Bengali).
2. Forms should be intuitive and accessible even to semi-literate users.
3. Icons and tooltips should be used for better understanding.

Performance

1. Login, submission, and loading time should not exceed 3 seconds.
2. The system must handle up to 10,000 concurrent users without lag.

Security

1. Data such as NID, phone number, and bank info must be encrypted at rest and in transit.
2. Role-based access control will limit unauthorized access.
3. Session timeout after 30 minutes of inactivity.

Reliability

1. System must achieve 99.9% uptime during peak harvest and loan application seasons.
2. Real-time backups should be performed every 6 hours.
3. OTP for password recovery must expire within 5 minutes.

Scalability

1. The system should support expansion to new districts or regions with minimal configuration changes.
2. New features like weather forecasts or agri-advisory services should be easily integratable.

Auditability

1. All transactions and requests should be logged with timestamps and user IDs.
2. Admins can generate weekly and monthly audit reports.

Availability

1. The mobile app must work in offline mode and sync when reconnected to the internet.
2. Web platform must be responsive and accessible on all modern browsers.

3.3. Software Development Constraints

Farmer Registration and Account Management

1. Registration must comply with national ID verification standards.
2. Admin interfaces should be accessible via desktop web browsers only.

Farmer Login

1. Login and authentication will use secure encryption protocols.
2. All login attempts must be logged for auditing and breach detection.

Loan/Subsidy and Lease Applications

1. Government approval workflows must be integrated into the admin panel.
2. Application status should reflect in real-time for transparency.

Driver Assignment

1. Location and route must integrate with Google Maps API.
2. Admins must track delivery progress and completion using timestamps.

Performance Limitations

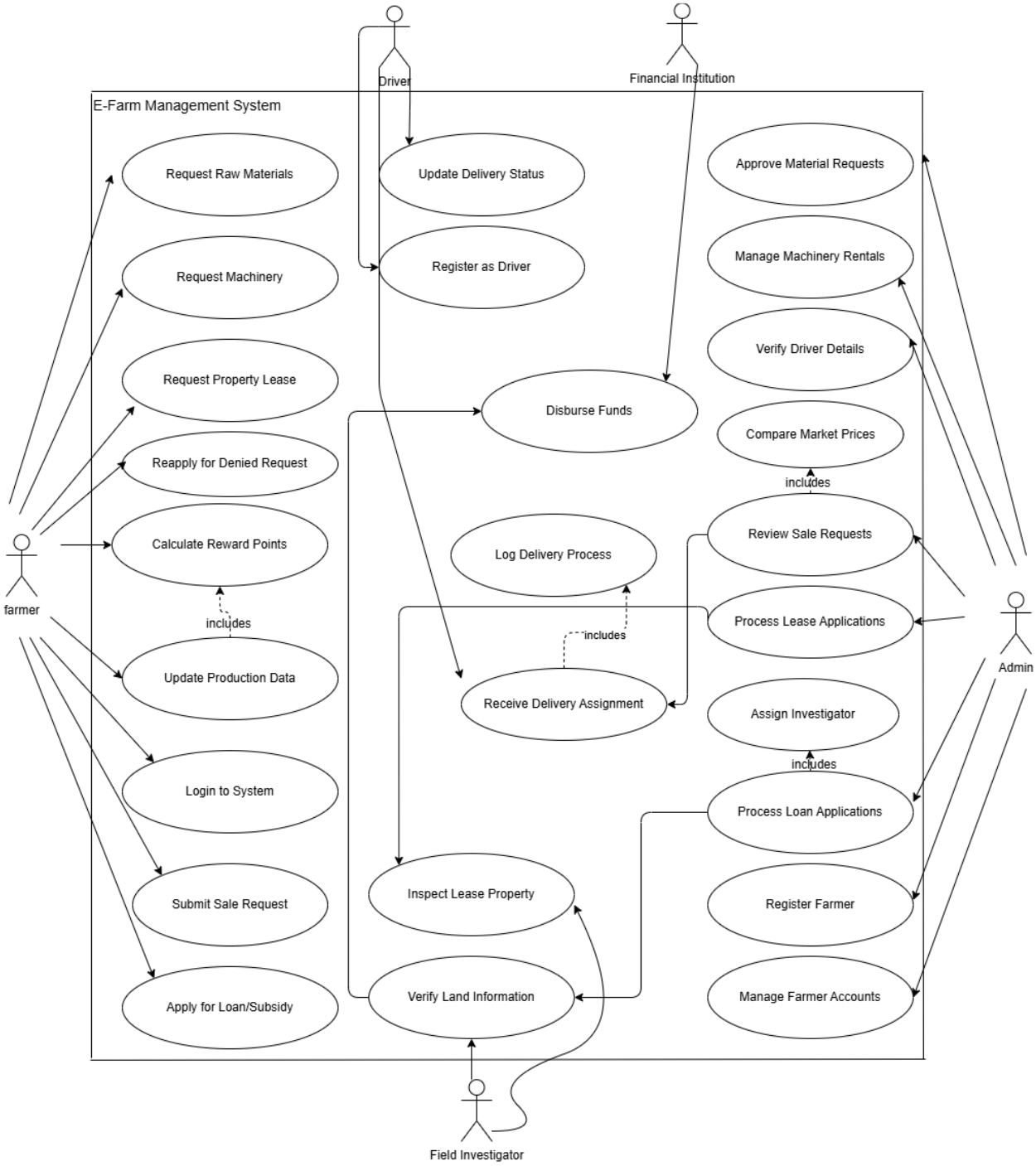
1. Budget limitations may require optimized resource use for hosting and backend infrastructure.
2. Offline support for farmers with poor connectivity will rely on caching and manual sync.

Compliance

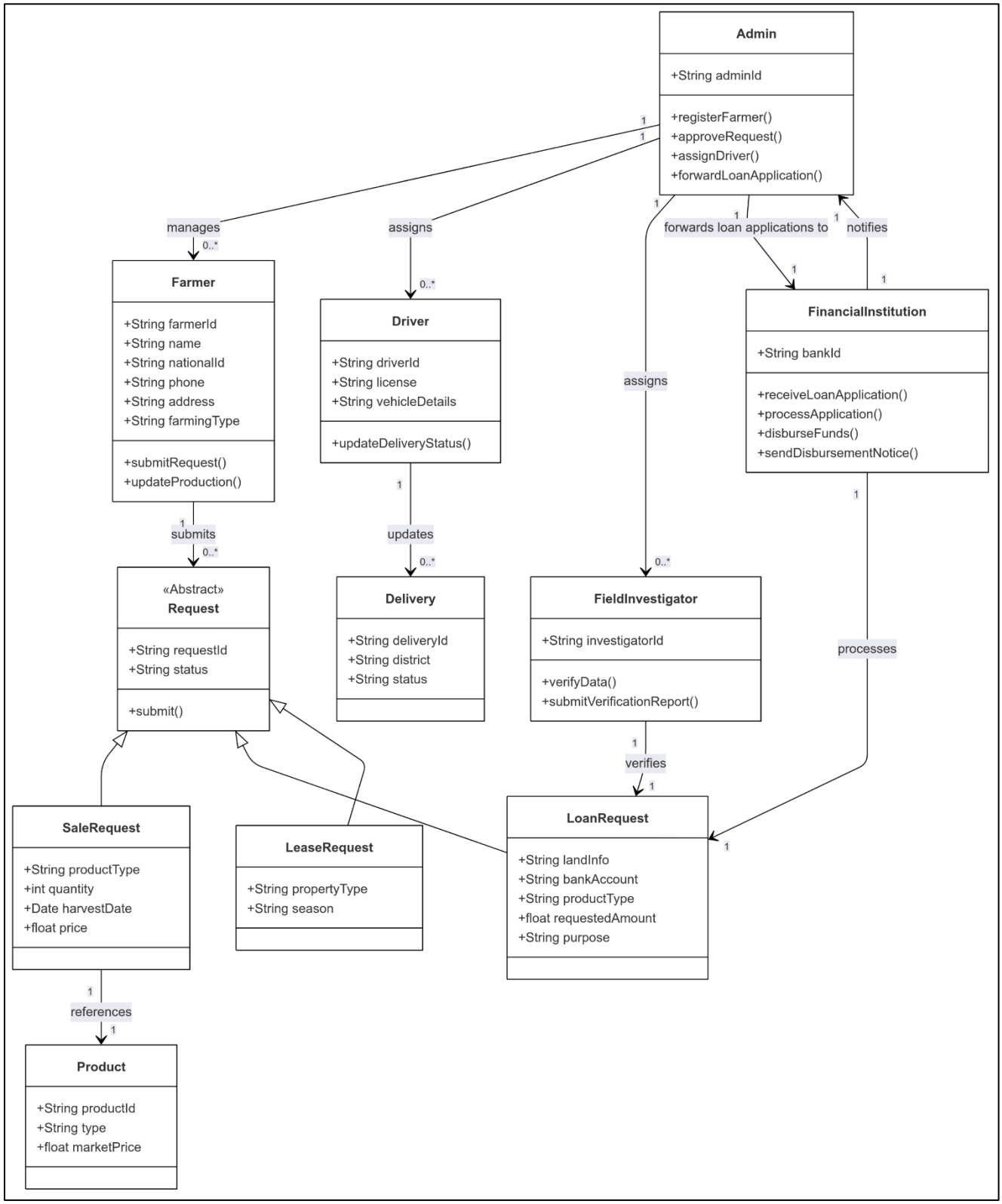
1. The system must follow local agriculture department regulations.
2. Farmer and financial data handling must comply with Bangladesh government digital security policies.

UML Diagrams:

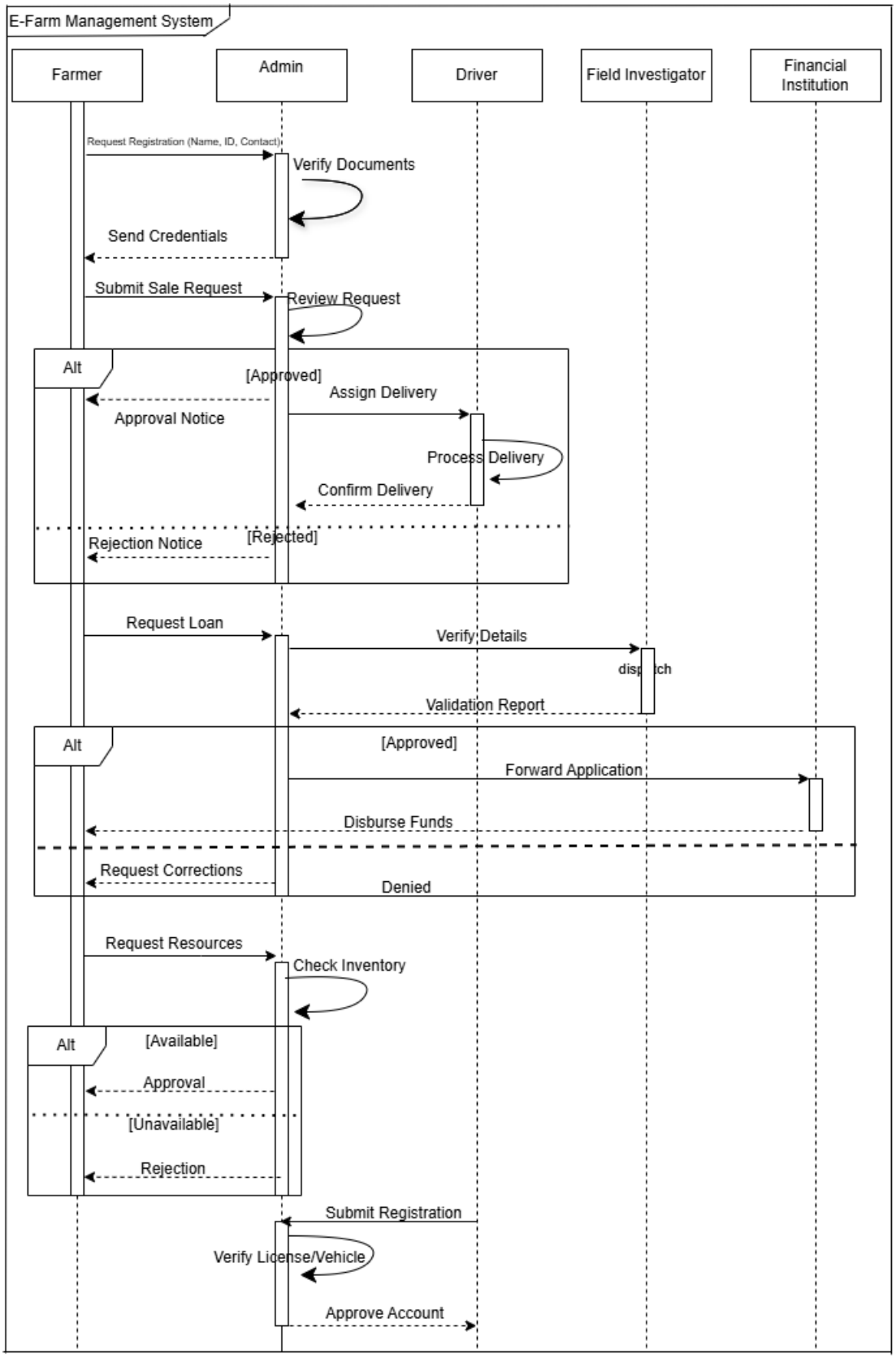
1. Use Case Diagram:



2. Class Diagram:

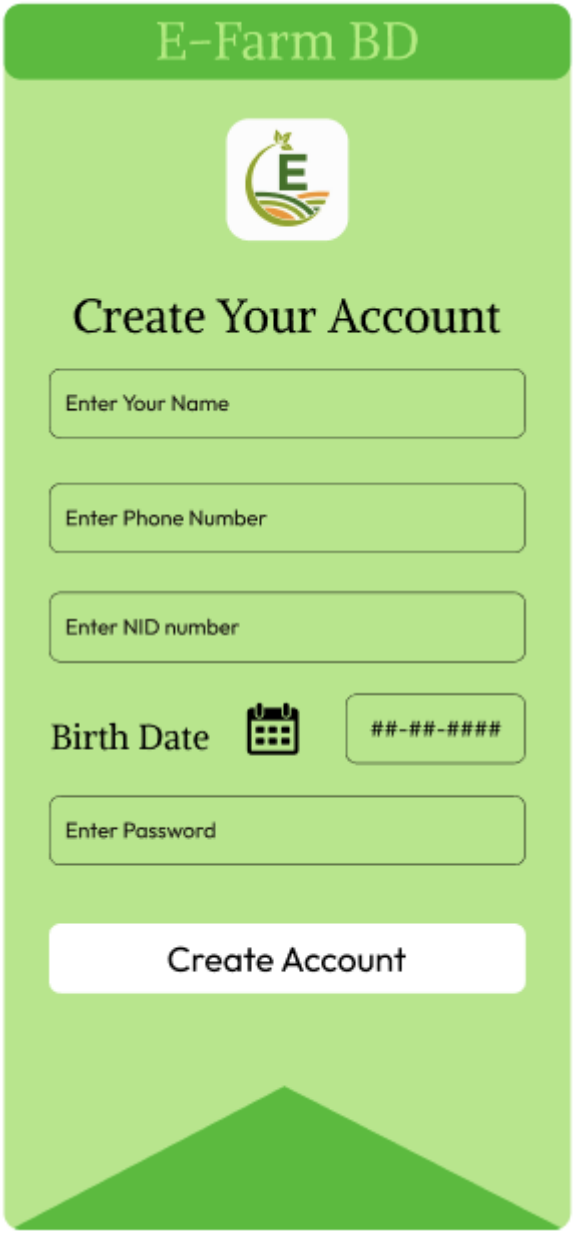


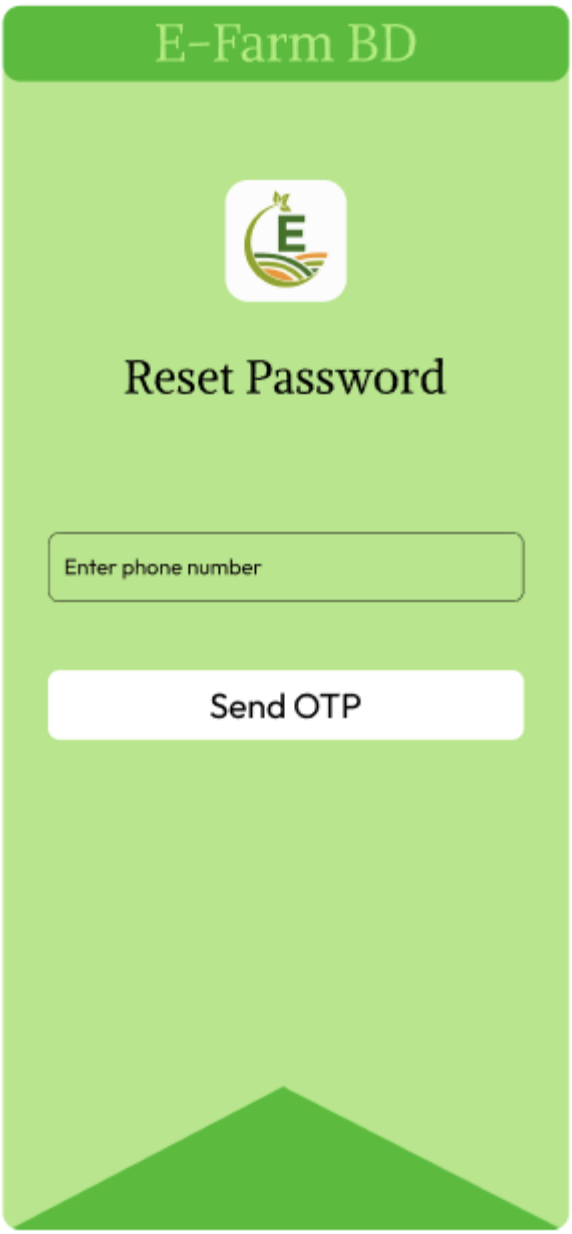
3. Sequence Diagram:

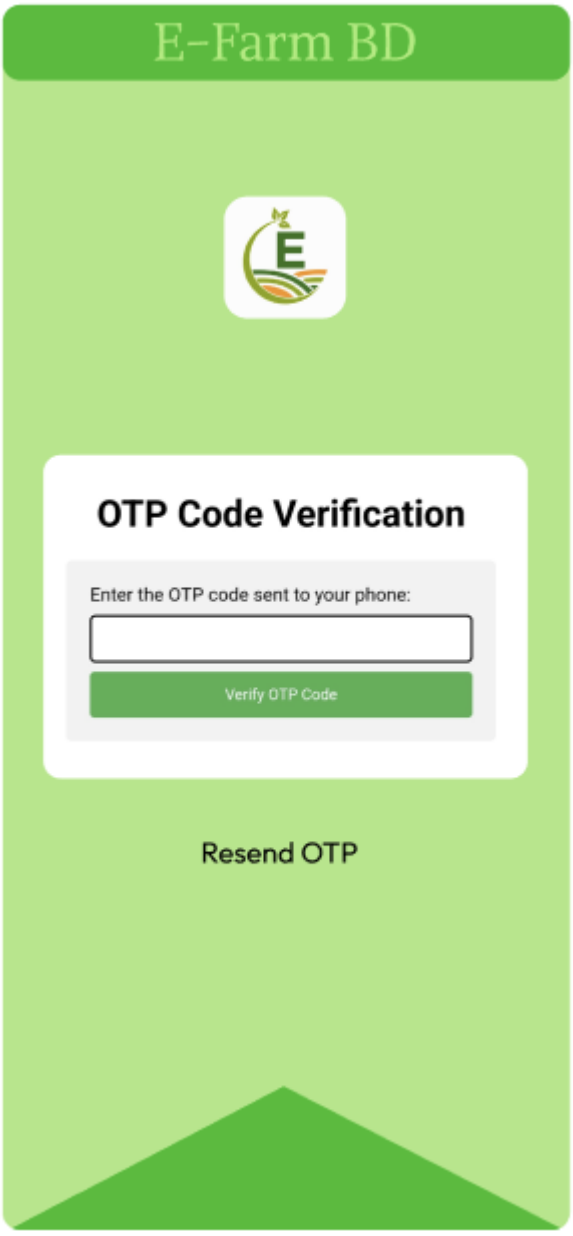


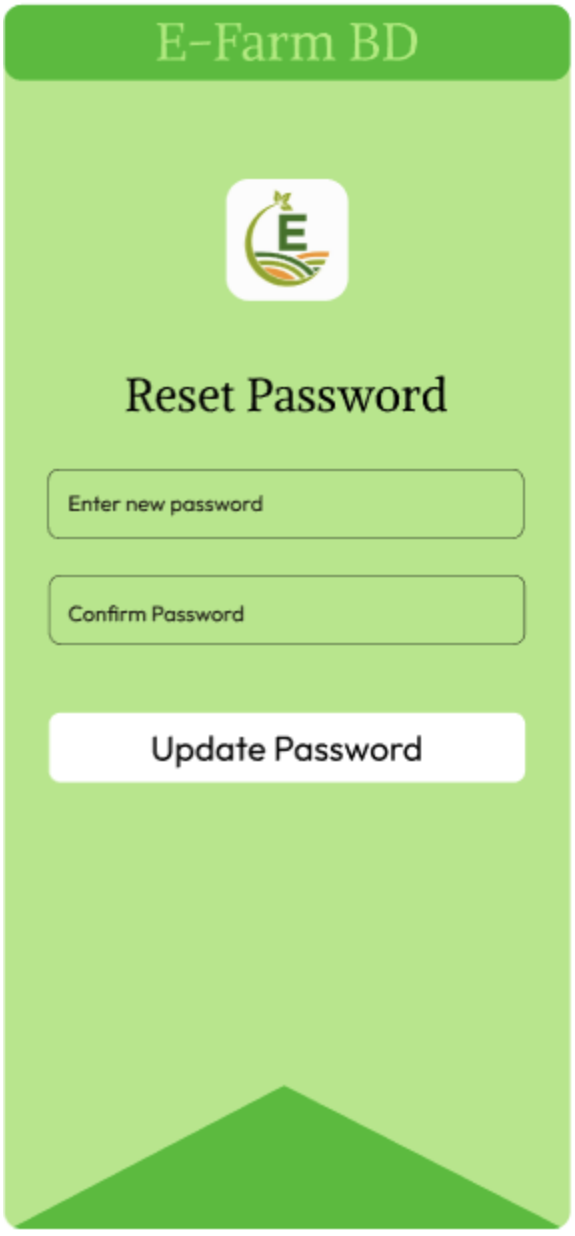
User Interface / User Experience (UI/UX):

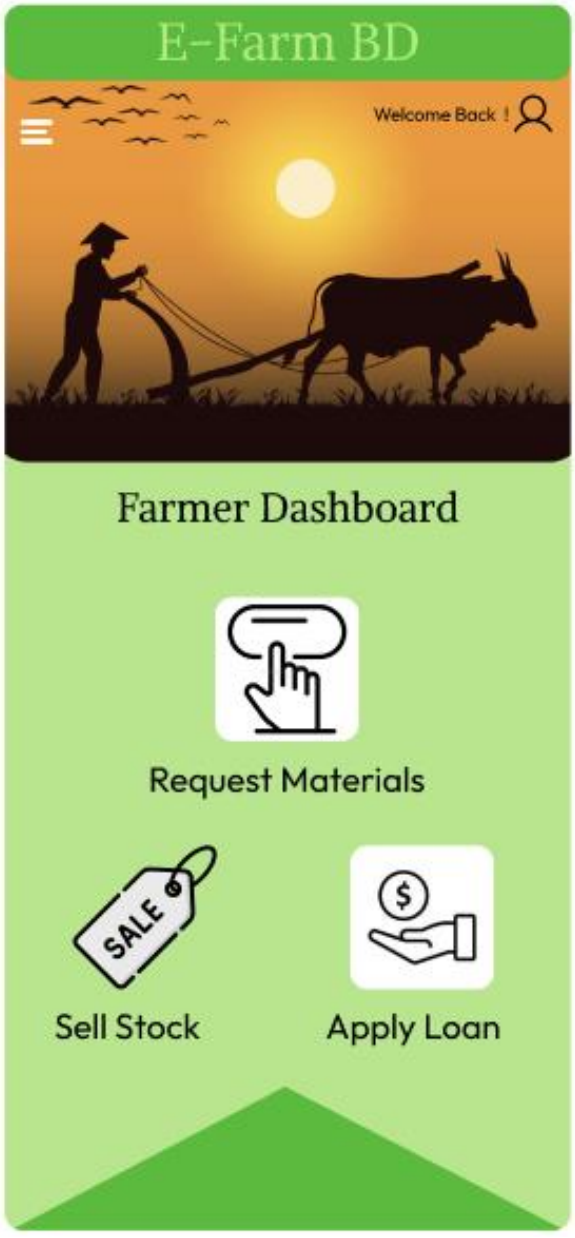




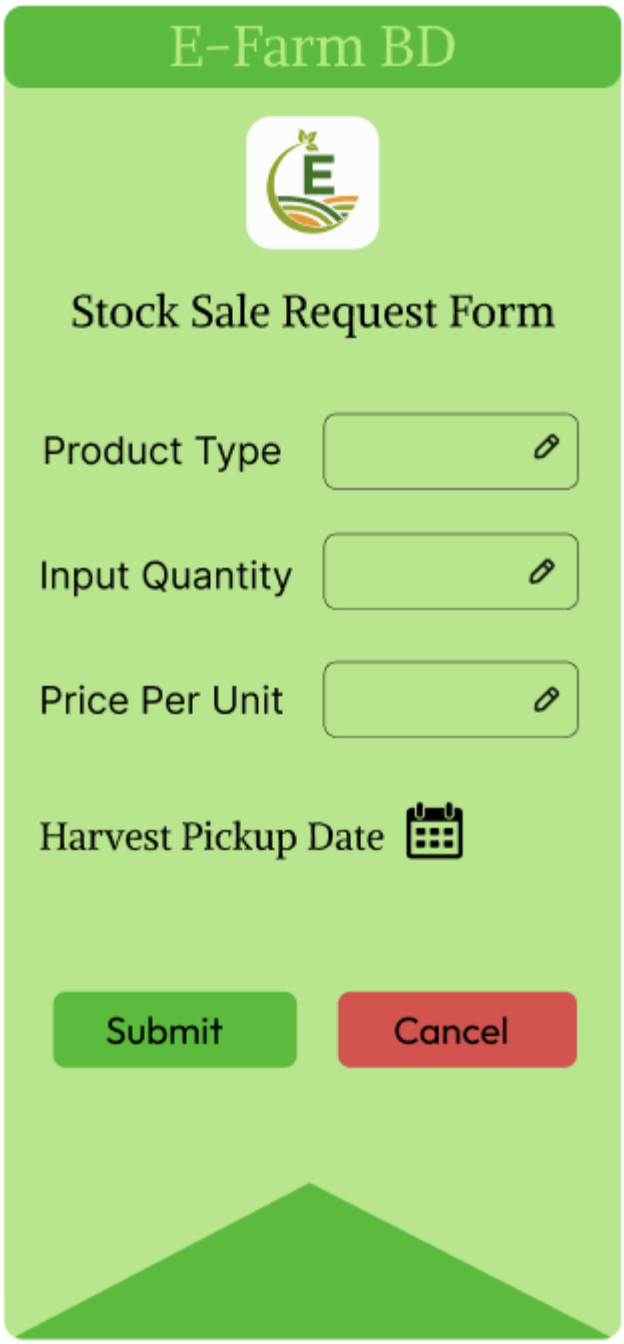

E-Farm BD
Reset Password
Enter phone number
Send OTP

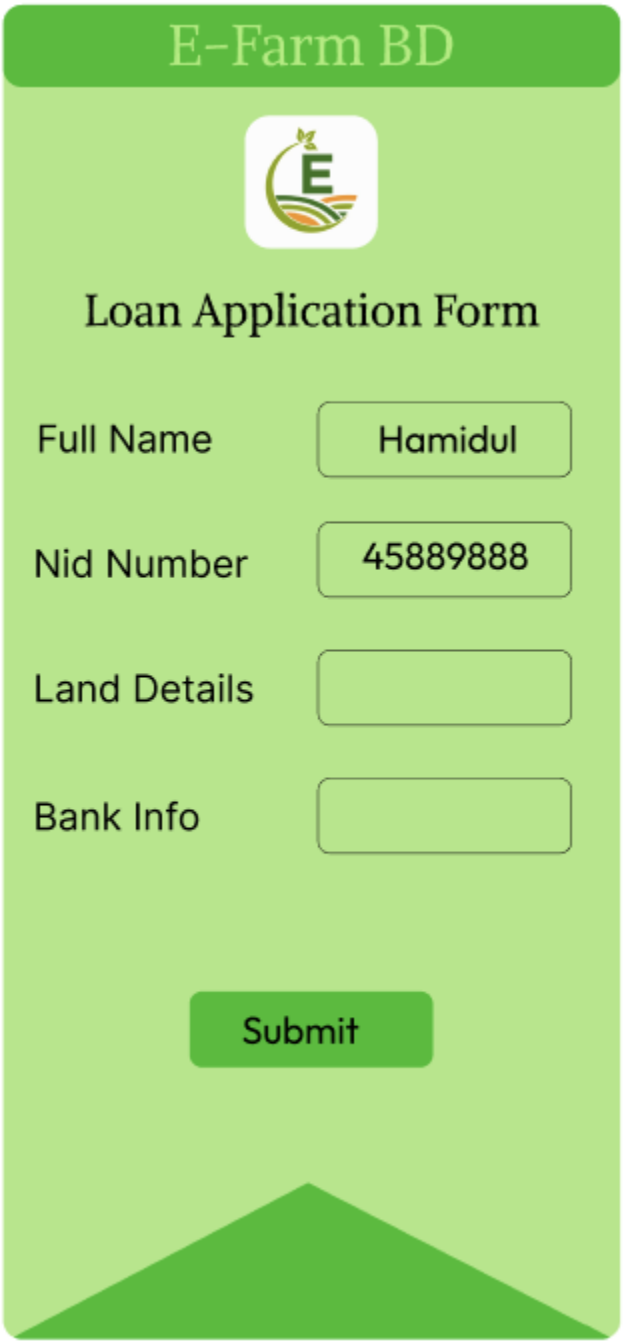

E-Farm BD
OTP Code Verification
Enter the OTP code sent to your phone:
Verify OTP Code
Resend OTP

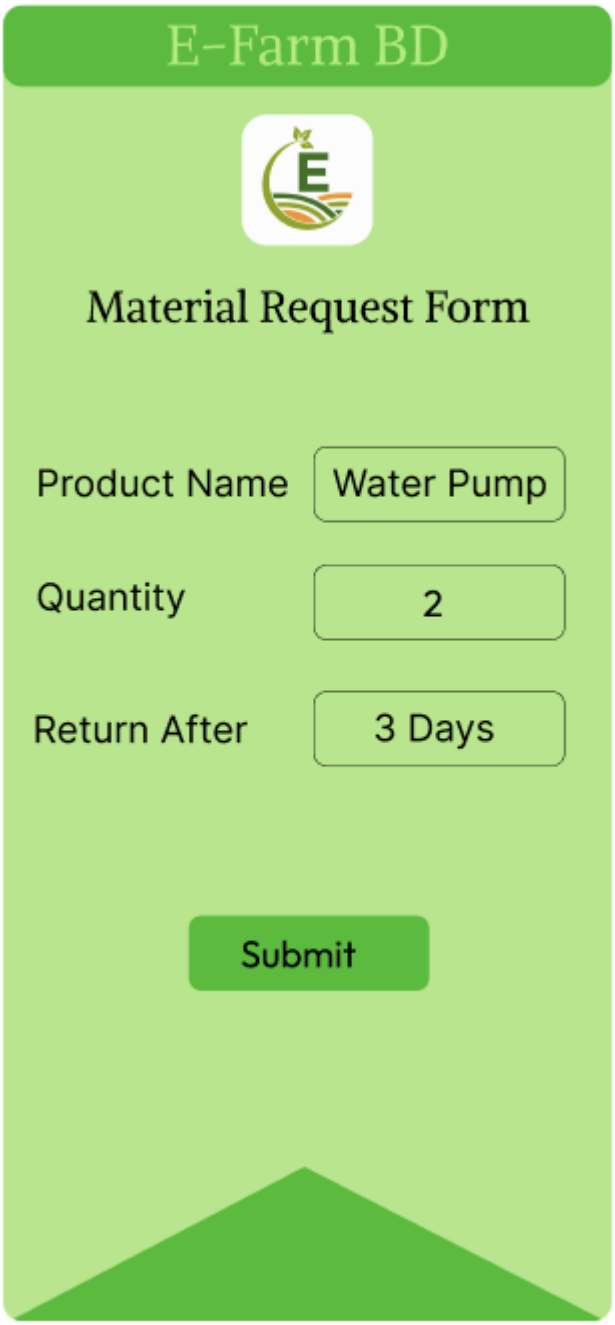

E-Farm BD
Reset Password
Enter new password
Confirm Password
Update Password

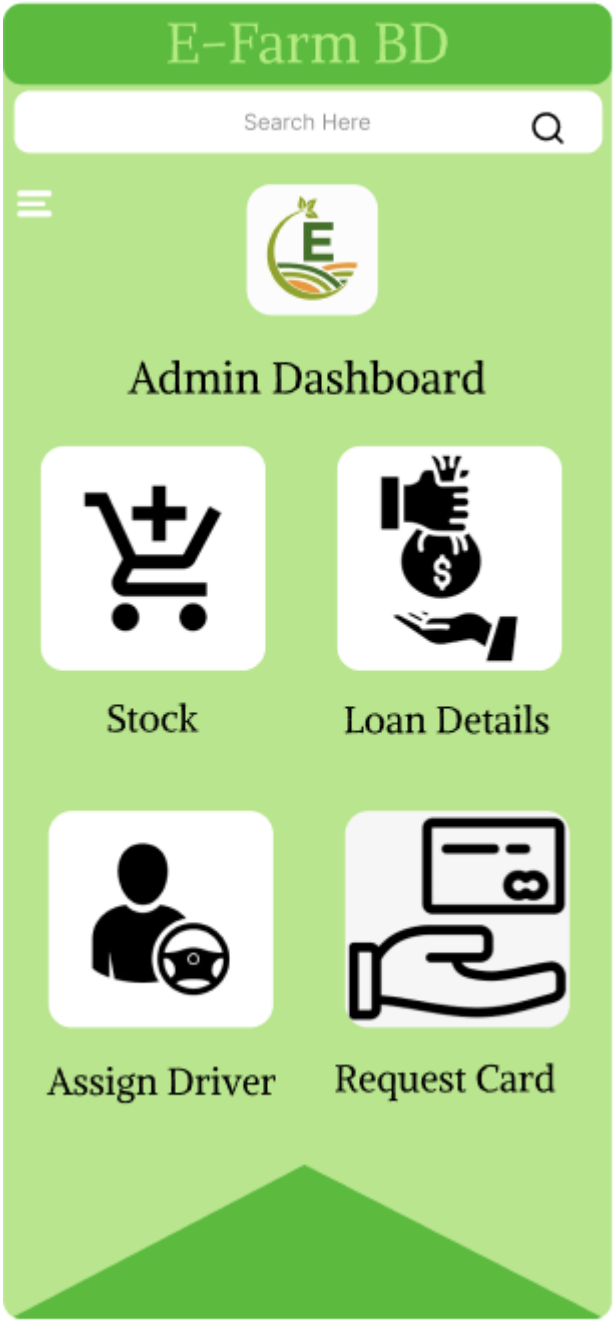

E-Farm BD
Welcome Back !
Farmer Dashboard
Request Materials
SALE
Sell Stock
Apply Loan

© MMH


E-Farm BD
Stock Sale Request Form
Product Type
Input Quantity
Price Per Unit
Harvest Pickup Date
Submit
Cancel


E-Farm BD
Loan Application Form
Full Name
Hamidul
Nid Number
45889888
Land Details
Bank Info
Submit


E-Farm BD
Material Request Form
Product Name
Water Pump
Quantity
2
Return After
3 Days
Submit


E-Farm BD
Search Here
Admin Dashboard
Stock
Loan Details
Assign Driver
Request Card

© MMH


E-Farm BD
Farmer Request Card
Farmer Name
Shahadat
Product Name
Tractor
Status ------ Pending!!
Approve
Reject


E-Farm BD
Pending Loan
Farmer Name
Land Details
Status ------ Pending!!
Approve
Reject

E-Farm BD

Assign Driver For Delivery

Driver Name Mostafizur

Vehicle Number ab-874

Status Available



Pick Up



Drop

Assign

E-Farm BD



Driver Dashboard

Pending Delivery

Completed Delivery

Test Modeling Use Case

The test cases in our project focus on the key features of the **E-Farm Management System** that ensure a smooth and efficient experience for users. The **registration and login** features securely verify user information, allowing easy access to the system. Farmers can easily **list and manage their products**, while real-time **inventory updates** help track availability. The system also includes a **driver management feature** that helps assign and track deliveries efficiently. These test cases prove that the system is reliable, easy to use, and supports the goal of improving digital farming services for local communities.

1. Test Case: Farmer Account Control System

Project Name: E-Farm Management System	Test Designed by: Mostafizur Rahman
Test Case ID: 1	Test Designed date:09-06-2025
Test Priority (Low, Medium, High): High	Test Executed by:
Module Name: Farmer Account Control System	Test Execution date:
Test Title: Farmer Registration and Account Control by Admin	
Description: Verifies that the admin can register, update, or delete farmer accounts with valid details and system performs necessary verifications.	
Precondition (If any): Admin is logged into the system with valid credentials	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to "Register Farmer" page 2. Enter full name, NID, phone number, address, and farming type 3. Submit the form 4. System verifies NID and phone number 5. Credentials are sent to the farmer	Full Name: "Rahim Uddin" NID: "1998456721345" Phone: "017xxxxxxxx" Address: "Kurigram, Bangladesh" Farming Type: "Rice"	Farmer is successfully registered and receives credentials. System allows updating or deleting account as needed.		
Post condition: Farmer account created/updated/deleted; credentials securely sent to farmer				

2. Test Case: Farmer Login

Project Name: E-Farm Management System			Test Designed by: Mostafizur Rahman	
Test Case ID: 2			Test Designed date:09-06-2025	
Test Priority (Low, Medium, High): High			Test Executed by:	
Module Name: Farmer Login			Test Execution date:	
Test Title: Farmer Login and Credential Verification				
Description: Verifies that farmers can log in using valid credentials, view the dashboard, and recover passwords if needed.				
Precondition (If any): Farmer is already registered				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open login page 2. Enter phone number and password 3. Click "Login" 4. Try invalid login 5 times 5. Use "Forgot Password" with phone/NID	Phone: "017xxxxxxx" Password: "Farm@1234" NID: "198456721345"	Valid login: Dashboard shown 5 wrong attempts: Account locked 1 hour Password recovery works		
Post Condition: Farmer is either logged in with dashboard access or temporarily locked after fails				

3. Test Case: Recover Password

Project Name: E-Farm Management System			Test Designed by: Mostafizur Rahman	
Test Case ID: 3			Test Designed date:09-06-2025	
Test Priority (Low, Medium, High): Medium			Test Executed by:	
Module Name: Password Recovery			Test Execution date:	
Test Title: Farmer Password Reset via NID or Phone				
Description: Ensures that farmers can reset password using NID or phone, and system enforces OTP and password policy.				
Precondition (If any): Farmer has valid NID or registered phone number				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open "Forgot Password" 2. Enter NID or phone 3. Submit and receive OTP 4. Enter OTP 5. Set strong new password 6. Confirm reset	Phone: "017xxxxxxxx" NID: "1998456721345" OTP: "123456" New Password: "NewPass@123"	OTP is verified, password is updated, SMS confirmation sent		
Post Condition: Password reset successfully and confirmation SMS sent to farmer				

4. Test Case: Stock Sale Request Management

Project Name: E-Farm Management System			Test Designed by: Shahadat Hossain	
Test Case ID: 4			Test Designed date:11-06-2025	
Test Priority (Low, Medium, High): High			Test Executed by:	
Module Name: Stock Sale Request Management			Test Execution date:	
Test Title: Farmer Stock Sale Request and Admin Approval				
Description: Verifies that farmers can submit stock sale requests and admins can approve/reject them based on criteria.				
Precondition (If any): Farmer is logged in with a valid account				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to sale request form 2. Enter product, quantity, price, harvest date 3. Submit request 4. Admin reviews & approves/reject 5. Check if pickup is scheduled 6. Edit/cancel before approval	Product: "Potato" Quantity: 500 kg Price: 20BDT/kg Harvest Date: "2025-07-10"	Requests are submitted, admin acts, and system responds accordingly		
Post Condition: Sale request is recorded, approved/rejected, or cancelled before approval				

5. Test Case: Loan and Subsidy Application

Project Name: E-Farm Management System	Test Designed by: Shahadat Hossain
Test Case ID: 5	Test Designed date:11-06-2025
Test Priority (Low, Medium, High): High	Test Executed by:
Module Name: Loan and Subsidy Application	Test Execution date:
Test Title: Farmer Loan and Subsidy Application and Processing	
Description: Verifies that farmers can apply for financial support and system handles validation, approval, and resubmission.	
Precondition (If any): Farmer is logged in and eligible to apply	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1.Open application form 2.Enter bank, land, crop, and amount 3.Submit for admin validation 4.Admin forwards for inspection 5.If approved, fund is transferred 6.Resubmit if denied	Bank: “Agrani Bank” Land: “2 acres” Crop: “Jute” Amount: “50,000 BDT”	Application is validated, inspected, approved, and funds are transferred to account		
Post Condition: Application saved, forwarded for inspection, or resubmitted if denied				

6. Test Case: Raw Material and Machinery Request

Project Name: E-Farm Management System			Test Designed by: Shahadat Hossain	
Test Case ID: 6			Test Designed date:11-06-2025	
Test Priority (Low, Medium, High): High			Test Executed by:	
Module Name: Raw Material and Machinery Request			Test Execution date:	
Test Title: Request and Approval of Materials and Machinery				
Description: Verifies that farmers can request raw materials or machinery, and system handles approval, subsidy, and rental logging.				
Precondition (If any): Farmer is logged in and eligible for request				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to request page 2. Select material/machinery 3. Enter quantity or rental period 4. Submit request 5. Admin approves/denies 6. Check subsidy/rental history	Item: "Fertilizer" Qty: "200 kg" Machine: "Tractor" Rental Period: "1 month"	Request is processed, subsidy applied if eligible, and rental is logged		
Post Condition: Request logged; subsidy/rental applied and recorded in farmer history				

7. Test Case: Production and Sale Update

Project Name: E-Farm Management System			Test Designed by: Hamidul Islam	
Test Case ID: 7			Test Designed date:12-06-2025	
Test Priority (Low, Medium, High): Medium			Test Executed by:	
Module Name: Production and Sale Update			Test Execution date:	
Test Title: Harvest Record Entry and Activity Scoring				
Description: Verifies that farmers can submit harvest details, admins validate them, and the system updates farmer activity scores.				
Precondition (If any): Farmer is logged in and has recent harvest to report				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to update page 2. Enter product type, date, quantity, price, market 3. Submit record 4. Admin verifies entry 5. System updates activity score	Product: “Wheat” Date: “2025-06 01” Qty: “300 kg” Price: “24 BDT/kg” Market: “Bogura Bazar”	Entry is saved, validated by admin, and activity score is updated		
Post Condition: Entry validated by admin and activity score updated				

8.Test Case: Farmer Priority System

Project Name: E-Farm Management System			Test Designed by: Hamidul Islam	
Test Case ID: 8			Test Designed date:12-06-2025	
Test Priority (Low, Medium, High): Medium			Test Executed by:	
Module Name: Farmer Priority System			Test Execution date:	
Test Title: Tier Ranking Based on Activity and Transparency				
Description: Verifies that farmers are ranked based on performance, and higher tiers receive prioritized services with visible tier status.				
Precondition (If any): Farmer has submitted valid data and maintained consistent activity				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Submit valid production/sale data 2. Maintain regular system use 3. Check dashboard for tier 4. Request a service 5. Verify prioritization	Activity Score: 85 Transparency: High Tier Expected: Gold	Farmer is ranked correctly, tier is shown on dashboard, and services are prioritized		
Post Condition: Tier updated based on recent farmer data/activity and visible on dashboard				

9. Test Case: Government Land or Pond Lease Application

Project Name: E-Farm Management System	Test Designed by: Hamidul Islam
Test Case ID: 9	Test Designed date:12-06-2025
Test Priority (Low, Medium, High): Medium	Test Executed by:
Module Name: Government Lease Application	Test Execution date:
Test Title: Application and Approval of Land /Pond Lease	
Description: Verifies that farmers can apply for government-owned land or pond lease, followed by inspection and official certification.	
Precondition (If any): Farmer is logged in and eligible to apply	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open lease application page 2. Select land/pond type 3. Submit required details 4. Admin assigns investigator 5. Inspection is done 6. Approval and lease certificate issued	Type: “Pond” Location: “Rangpur” Purpose: “Fish farming”	Application is reviewed, verified, and approved with certificate issued		

Post Condition: Lease request approved with inspection and certificate issued

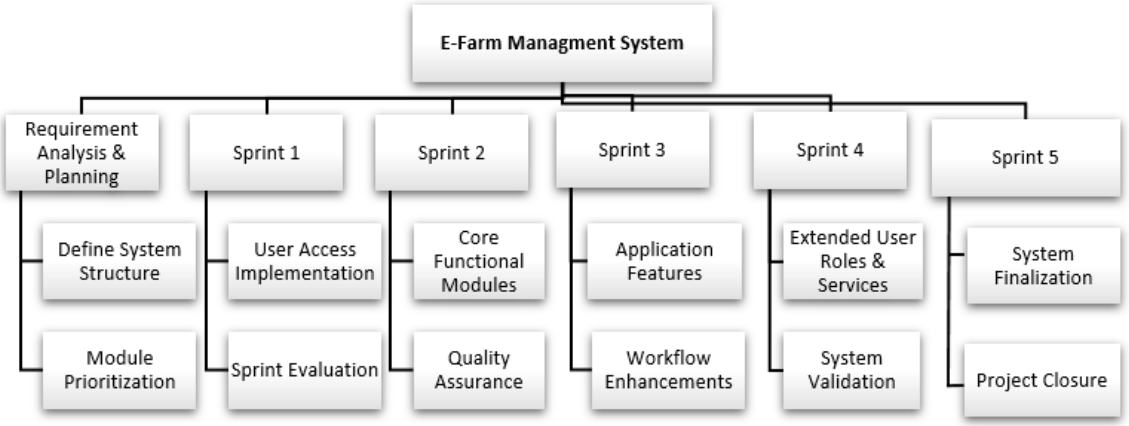
10.Test Case: Driver Registration and Delivery Assignment

Project Name: E-Farm Management System	Test Designed by: Mostafizur Rahman
Test Case ID: 10	Test Designed date:12-06-2025
Test Priority (Low, Medium, High): High	Test Executed by:
Module Name: Driver Registration and Delivery Assignment	Test Execution date:
Test Title: Driver Registration, Approval, and Task Assignment	
Description: Verifies that drivers can register, admins approve them, and deliveries are assigned with proper wage calculations.	
Precondition (If any): Driver has valid documents and is not already registered	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open driver registration form 2. Enter NID, license, phone, vehicle info 3. Submit form 4. Admin reviews and approves 5. Admin assigns delivery 6. Verify wage based on distance/load	NID: "1998567890123" License: "DL-450932" Vehicle: "Pickup Van" Destination: "Khulna"	The driver is registered, approved, assigned delivery, and paid with incentives		

Post Condition: Driver is approved, assigned delivery, and wage calculated based on task

Work Breakdown Structure (WBS):



COCOMO Calculation:

Estimation Basis:

- Project scope includes multiple complex modules (registration, loan system, admin panel, driver logistics)
- **SLOC (Estimated) = 7000 lines**
- **COCOMO model**: Organic (simple project, small team, well-understood domain)
- **Coefficient** = 2.4, **P** = 1.05

PM: person-months needed for project (labor working hours)

PM = Coefficient$_{\text{<Effort Factor>}}$*(SLOC/1000)^P

= 2.4*(7000/1000)^1.05

= 18.517

DM: duration time in months for project (week days)

DM = 2.50*(PM)^T

=2.50*(18.517)^0.38

= 7.579 Months

ST: average staffing necessary

ST = PM/DM = 18.517/7.579 = 2.443 ~ **3 team members**

Gantt Chart:

HI

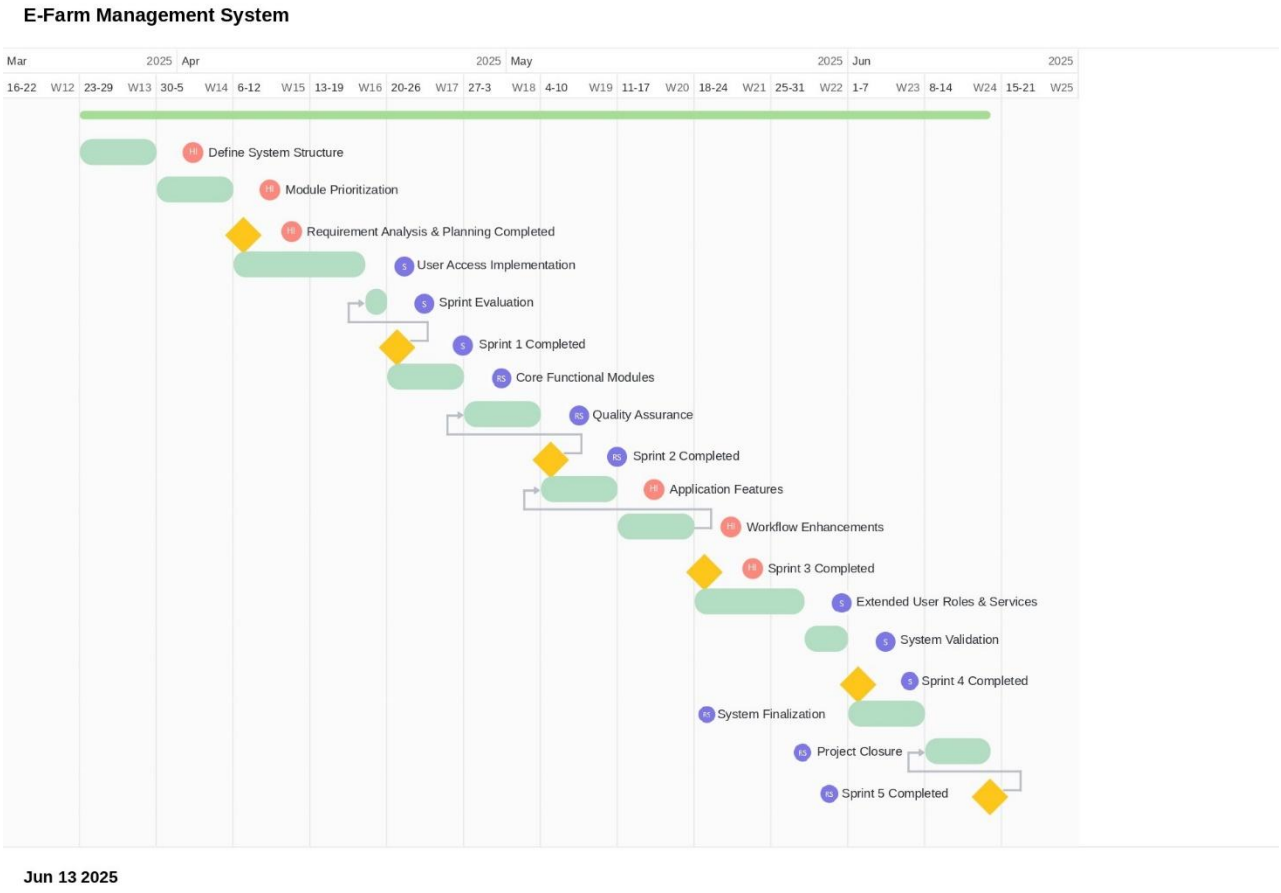
Represent **Hamidul Islam**

S

Represent **Shahadat Hossain**

RS

Represent **Mostafizur Rahman**



EVA Calculation:

Task	Planned	Actual Effort
1. Requirement Planning	6	6.5
2. User Access & Sprint Evaluation	7	8
3. Core Functional Modules	8	9
4. Quality Assurance	7	6
5. Application Features	8	8.5
6. Workflow Enhancements	10	11
7. Extended Roles & Services	10	12
8. System Validation	9	9.5
9. System Finalization	10	10.5
10. Project Closure	10	11
Total Task = 10	**BCWP = 51** **BCWS = 85**	**ACWP = 91**

- **BCWP (Earned Value)** = Assumed 60% of total work has been completed:
  BCWP = 60% of total planned effort = 0.6 × 85 = 51 person-days
- **BAC (Budget at Completion) =Estimated total = 300**
- **SPI = BCWP / BCWS = 51 / 85 = 0.600**
- **SV = BCWP – BCWS = 51- 85 = -34 person-days**
- **CPI = BCWP /ACWP = 51 / 91 = 0.560**
- **CV = BCWP -ACWP = 51 – 91 = -40 person-days**

% schedule for completion = **BCWS / BAC =** $\frac{85}{300} \times$ **100 = 28.33%**

[% of work scheduled to be done at this time]

% Work complete = **BCWP / BAC =** $\frac{51}{300} \times$ **100 = 17.00%**

[% of work completed at this time]

11. Risk Management Plan

This document presents a comprehensive risk management plan tailored for the **E-Farm Management System**. The purpose of this plan is to identify, assess, and mitigate potential risks that may affect the successful development, deployment, and operation of the system. Given the project's societal impact, technical complexity, and multi-stakeholder nature, proactive risk management is essential to ensure sustainability and reliability.

Risk Identification and Assessment

The following table identifies key risks associated with the E-Farm Management System, categorized by type, along with their probability, impact, and mitigation strategies:

Risk	Category	Probability (%)	Impact (1–3)	Risk Mitigation Strategy
Low digital literacy among rural farmers	Customer	70%	3	Provide Bengali-language support, icon-based UI, and offline features. Offer in-person training via local admins.
Delay in integration of modules due to limited team size	Project Schedule	60%	3	Use Scrum sprints and assign parallel tasks to balance workloads. Monitor progress weekly.
Server overload during peak farming seasons	Technology	50%	3	Use scalable cloud infrastructure with load balancing and regular stress testing.
Inaccurate or fake data entries by users (e.g., crop data, loan requests)	Data Integrity	40%	3	Implement NID validation, audit trails, and manual admin verification for sensitive requests.
Lack of stakeholder engagement from government or local offices	Stakeholder	30%	3	Schedule regular feedback meetings and engage stakeholders through early demos and workshops.
Security breaches involving farmers or financial data	Security	40%	3	Apply strong encryption, role-based access control, and periodic security audits.
Unreliable internet	Technology	60%	2	Enable offline data entry with background sync.

connectivity in rural areas				Ensure lightweight mobile UI.
Conflicts between stakeholders over resource allocation or priorities	Management	30%	2	Maintain transparent logs, use priority scoring system, and offer admin override capabilities.
Delays in receiving transport availability updates from drivers	Operational	50%	2	Set notification reminders for drivers, include real-time status update requirements, and allow admins to manually reassign tasks.
Resistance to platform adoption from traditional farmers	Customer Behavior	50%	2	Launch awareness campaigns and reward early adopters using the priority tier system.
Bugs discovered in critical modules	Quality	40%	3	Conduct frequent sprint-based testing, including unit and integration tests after each deliverable.

Risk Monitoring and Control

Risk monitoring will be a continuous activity throughout the project lifecycle. The team will maintain a dynamic **risk register**, review it at the end of each sprint, and adjust mitigation strategies as needed. Specific sprint retrospectives will address resolved and new risks, and priority will be given to high-impact items. Documentation, weekly progress meetings, and regular backup procedures will support this control framework.

Conclusion:

The E-Farm Management System is a comprehensive and socially impactful digital solution designed to address long-standing inefficiencies in Bangladesh's agricultural sector. By digitizing farmer registration, crop sales, subsidy and loan applications, and transport logistics, the system empowers farmers while enhancing transparency and operational efficiency for government agencies. The adoption of a hybrid Incremental and Scrum development model has enabled modular, flexible, and timely delivery of system components, allowing the team to adapt to evolving requirements and academic deadlines. With a strong focus on usability, security, and scalability, the E-Farm Management System lays a solid foundation for future expansion into areas like IoT-based crop monitoring and data-driven agricultural policymaking. This project not only demonstrates sound software engineering principles but also contributes meaningfully to solving real-world problems that affect millions of lives.

Rubric for Project Assessment (CO3)

Criteria	Marks distribution (Max 3X5= 15)				Acquired Marks
	Inadequate (1-2)	Satisfactory (3)	Good (4)	Excellent (5)	
Selection of Software Engineering Models	Does not articulate a position or argument of choosing appropriate model. Does not present any evidence to support the arguments for the choice of the model	Articulates a position or argument for choosing models that is unfocused or ambiguous. Presents incomplete/vague evidence to support argument for model choice	Articulates a position or argument of choosing models that is limited in scope. Does not present enough evidence to support the argument for the choice of the model	Clearly articulates a position or argument for the choosing software engineering models. Presents sufficient amount of evidence to support argument for the model selection	
Role identification and Responsibility Allocation	The project has poor project management plans for identifying roles and assigning the responsibilities	Identify few roles in the project management where some of the roles are left alone with any project responsibilities	Identify most of the roles in the project management and assign their responsibilities	Well planned project with proper role identification and responsibility allocation in the project management activities	
Impact identification					
Formatting and Submission	Project report is not complete and Several errors in spelling and grammar. Present a Confusing	Some errors in spelling and grammar. Some problems	Few errors in spelling and grammar. Presents most of the details in	Project report is complete and No errors in spelling and grammar. Consistently	

	organization of concepts, supporting arguments, and real-life example. Sentences rambling, and details are repeated.	of organizing the answer in a logical order of defining, elaborating, and providing real-life examples.	a logical flow of organization in definition, details, and example.	presents a logical and effective organization of definition, details, and real-life example of the topic.	
				Acquired marks:	
				CO Pass / Fail:	

Rubric for Project Assessment (CO4)

Marking Criteria	Marks Distribution (Maximum 3X5=15)				Acquired Marks
	Inadequate (1-2)	**Satisfactory (3)**	**Good (4)**	**Excellent (5)**	
Project Planning	No background information regarding the project is given; project goals and benefits are missing.	Insufficient background information is given; project goals and benefits are poorly stated	Sufficient background information is given; the purpose and goals of the project are explained.	Thorough and relevant background information is given; project goals are clear and easy to identify.	
Effort Estimation and Scheduling	Student vaguely discuss the impact of societal, health, safety, legal and cultural issues in their project	Student provided with partial relevance to the impact of societal, health, safety, legal and cultural issues in their project	Student fairly provided the analysis to the impact of societal, health, safety, legal and cultural issues in their project	Student comprehensively provided the analysis to the impact of societal, health, safety, legal and cultural issues in their project	
Risk Management	Ambiguous representative example.	Partially identify / indicate towards real-life example.	Real-life example is fairly connected towards the definition.	Comprehensively defend with real life example.	
				Acquired Marks:	
				CO Pass / Fail:	

CO5 [PO-i-2]: Perform as an effective team member or leader in diverse team settings and solve multi-disciplinary problems in computer science and engineering domain.				
Assessment Attribute/Criteria	**Missing/ Incorrect (0)**	**Inadequate (1)**	**Satisfactory (2)**	**Excellent (3)**
Taking responsibility	Does not perform assigned tasks; often misses meetings and, when present, does not have anything constructive to say; relies on others to do the work;	Partially performs all assigned tasks; attends meetings irregularly and occasionally participates and hence not reliable;	Performs all assigned tasks; attends meetings regularly and usually participates effectively. generally reliable;	Performs all tasks very effectively; attends all meetings and participates enthusiastically; very reliable.
Contributions	Never provides useful ideas when participating in a group discussion	Rarely provides useful ideas when participating in a group discussion	Sometimes provides useful ideas when participating in a group discussion	Routinely provides useful ideas when participating in a group discussion
Collaboration and Ability to Compromise	Not cooperative, unable to compromise and disrupts the team process.	Sometimes cooperative, and rarely displays a positive attitude.	Usually cooperative, able to compromise and generally display positive attitude.	Always cooperative. Willingness to compromise. Always display positive attitude.
Valuing other team members (Working with others)	Often argues with teammates; doesn't let anyone else talk; occasional personal attacks and "put-downs"; wants to have things done his way and does not listen to alternate approaches.	Seldom listens to others' points of view; occasionally behaves in an oppressive manner; tries to force their own ideologies on other.	Generally, listens to others' points of view; always uses appropriate and respectful language; tries to make a definite effort to understand others' ideas.	Always listens to others and their ideas; helps them develop their ideas while giving them full credit; always helps the team reach a fair decision.